

COMPSCI 589 Final Project - Spring 2026

Due **May 8**, 11:55 pm Eastern Time

Introduction

- Each group member was responsible for evaluating the performance of at least one unique algorithm implemented during the semester on each dataset
- The algorithms tested in this project were chosen from the pool of: k -NN, Decision Trees, multinomial Naive Bayes, Random Forests, and Neural Networks
- All work was done in the shared github repository https://github.com/1985lxw/589_final_project

-
- **Katherine** implemented TODO
 - **Tiffany** implemented TODO
 - **Xing-Wei** implemented Decision Trees

Experiments and Analyses

1 The Hand-Written Digits Recognition Dataset

1.1 Neural Networks Algorithm

TODO

1.2 Algorithm 2

TODO

1.3 Random Forest Algorithm

`\newpage`

`\section{\underline{The Oxford Parkinson's Disease Detection Dataset}}`

`\subsection{Neural Networks Algorithm}`

TODO

`\noindent\rule{\textwidth}{0.5pt}`

`\subsection{Algorithm 2}`

TODO

`\noindent\rule{\textwidth}{0.5pt}`

`\subsection{Random Forest Algorithm}`

`\newpage`

`\section{\underline{The Rice Grain Dataset}}`

`\subsection{Neural Networks Algorithm}`

TODO

`\noindent\rule{\textwidth}{0.5pt}`

`\subsection{Algorithm 2}`

TODO

\noindent\rule{\textwidth}{0.5pt}

\subsection{Decision Trees Algorithm}

We chose to test decision trees on the rice dataset because it has a moderate number of features (7) and a relatively large number of samples (3810), which allows us to test two different hyperparameters; tree depths and minimum split sizes. This is shown in the tuning results below, where we tested 6 different tree depths and 4 different minimum split sizes.

\begin{verbatim}

Tuning for dataset with 3810 rows and 7 features.

Testing Depths: [2, 3, 5, 7, 9, 11]

Testing Splits: [190, 381, 571, 762]

Depth	Split	Test Acc	Test F1

2	190	0.9215	0.9192
2	381	0.9228	0.9205
2	571	0.9226	0.9202
2	762	0.9239	0.9215
3	190	0.9234	0.9210
3	381	0.9213	0.9189
3	571	0.9202	0.9179
3	762	0.9239	0.9215
5	190	0.9213	0.9189
5	381	0.9234	0.9210
5	571	0.9226	0.9202
5	762	0.9239	0.9215
7	190	0.9223	0.9199
7	381	0.9226	0.9203
7	571	0.9215	0.9192
7	762	0.9239	0.9215
9	190	0.9234	0.9210
9	381	0.9220	0.9197
9	571	0.9239	0.9215
9	762	0.9239	0.9215
11	190	0.9223	0.9199
11	381	0.9220	0.9197
11	571	0.9218	0.9194
11	762	0.9239	0.9215

=====

BEST CONFIGURATION:

Max Depth: 11

Min Split: 762

F1 Score: 0.9215

=====

Final Results:

Training Accuracy: 0.9239

Training F1 Score: 0.9216

Testing Accuracy: 0.9239

Testing F1 Score: 0.9215

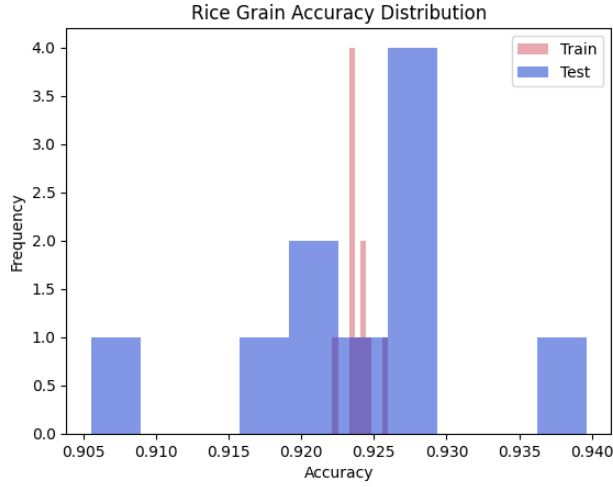


Figure 1: Rice Accuracy Distribution

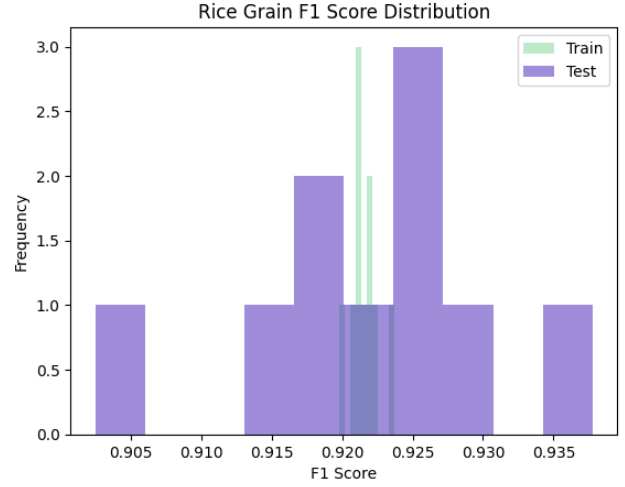


Figure 2: Rice F1 Score Distribution

For both models, the training scores are highly concentrated around 0.92, while the testing scores are more spread out, with a mean around 0.923 for accuracy and 0.922 for F1 score. The training performance suggests that the models have high stability and low bias, however, the testing performance indicates that there may be some sensitivity to the specific training data used in each fold, leading to a poor local optimum. Since the mean training scores are very close to the mean testing scores, it suggests that the models are robust and are not overfitting. The accuracy and F1 scores are also very close to each other which suggests that the models are performing well across all labels in the dataset and that the labels in the dataset are balanced. Decision trees can easily split numerical features to find precise boundaries, so they are well-suited for the rice dataset, which consists of numerical attributes.

2 The Credit Approval Dataset

2.1 Neural Networks Algorithm

TODO

2.2 Algorithm 2

TODO

2.3 Decision Trees Algorithm

We chose to test decision trees on the credit approval dataset because it has a moderate number of features (15) and a relatively small number of samples (653), which allows us to test two different hyperparameters; tree depths and minimum split sizes. This is shown in the tuning results below where we tested 6 different tree depths and 4 different minimum split sizes.

Tuning for dataset with 653 rows and 15 features.

Testing Depths: [2, 4, 6, 8, 10, 12]

Testing Splits: [32, 65, 97, 130]

Depth	Split	Test Acc	Test F1
2	32	0.8622	0.8621
2	65	0.8639	0.8636
2	97	0.8622	0.8622
2	130	0.8622	0.8620
4	32	0.8378	0.8358
4	65	0.8482	0.8473
4	97	0.8588	0.8580
4	130	0.8577	0.8566
6	32	0.8318	0.8293
6	65	0.8302	0.8288
6	97	0.8501	0.8491
6	130	0.8610	0.8606
8	32	0.8378	0.8356
8	65	0.8576	0.8568
8	97	0.8496	0.8489
8	130	0.8620	0.8612
10	32	0.8377	0.8356
10	65	0.8409	0.8397
10	97	0.8437	0.8423
10	130	0.8504	0.8495
12	32	0.8375	0.8350
12	65	0.8451	0.8439
12	97	0.8668	0.8654
12	130	0.8635	0.8626

=====
BEST CONFIGURATION:

Max Depth: 12

Min Split: 97

F1 Score: 0.8654
=====

Final Results:

Training Accuracy: 0.8746

Training F1 Score: 0.8741

Testing Accuracy: 0.8606

Testing F1 Score: 0.8601

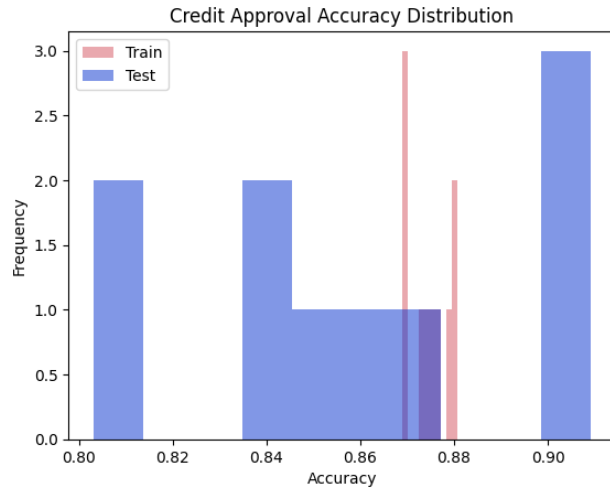


Figure 3: Credit Approval Accuracy Distribution

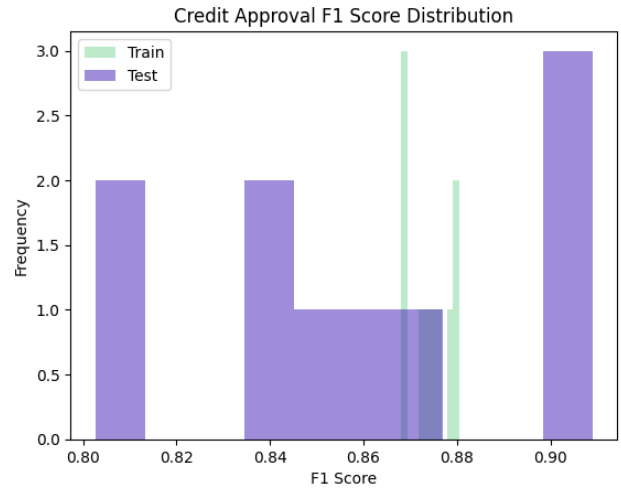


Figure 4: Credit Approval F1 Score Distribution

The test scores are widely spread which suggests the model is sensitive to the specific training data used in each fold, leading to a poor local optimum. Especially in smaller datasets, the presence/absence of a few instances in a training fold can have a significant impact on the tree structure and thus the performance. The training scores are more concentrated around 0.87, while some test folds outperform the training performance. This suggests that the model is stable during training, but is not consistent on generalizing depending on the test fold. The accuracy and F1 scores are also very close to each other which suggests that the model is performing well across all labels in the dataset and that the labels in the dataset are balanced. Since the credit approval dataset has some categorical features, they need to be factorized. If there are a lot of unique values in a categorical feature, it can lead to a large number of splits in the decision tree, which may be why there is more spread in the test scores for this dataset.

	Handwritten Digits		Parkinson's		Rice Grains		Credit Approval	
	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score
Neural Networks								
K-Nearest Neighbors								
Decision Trees	0.8344	0.8327	0.8508	0.8101	0.9239	0.9215	0.8606	0.8601

Extra Credit

We did the following extra credit tasks. Describe what you did as part of that task and present the corresponding results. Any source code you develop should be included in your Gradescope submission.

[QE.1] [Extra Credit Question #1] (10 Points)

You may earn up to 10 extra points if you analyze all four datasets using an additional algorithm. In other words, you would be evaluating more than two algorithms on each of the datasets discussed in Section 1.

[QE.2] [Extra Credit Question #2] (10 Points)

You may earn up to 10 extra points if you select a new challenging dataset and evaluate the performance of different algorithms on it. A challenging dataset should necessarily include both categorical and numerical attributes and have more than two classes; or it should be a dataset where you have to classify images.

[QE.3] [Extra Credit Question #3] (15 Points)

You may earn up to 15 extra points if you combine different algorithms and construct an ensemble. For instance, you could create an ensemble composed of three neural networks (each with a different architecture) and a random forest. The ensemble would be trained as described in class: each algorithm would be trained on its own bootstrap dataset, and so on. The final predictions would then be determined by majority voting. Evaluate your ensemble algorithm on all four datasets discussed in Section 1.

[QE.3] [Extra Credit Question #3] (15 Points)

You may earn up to 15 extra points if you implement a new type of algorithm—a non-trivial variant of one of the algorithms studied in class. If you choose to do so, you should evaluate its performance on all four datasets.